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Printing apparatus

Abstract

A printing apparatus includes: a printing unit that performs printing on a print medium; a support unit disposed facing the printing unit, the support unit supporting the print medium; a plurality of first transport rollers that transport the print medium to the support unit; a medium collection unit that winds in a roll shape the print medium on which printing was performed by the printing unit; and a plurality of second transport rollers that transport the print medium from the support unit to the medium collection unit. Each interval between the plurality of first transport rollers is narrower than an average value of each interval between the plurality of second transport rollers.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-129926, filed Aug. 17, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a printing apparatus.

2. Related Art

(3) In the related art, a printing apparatus with a steering unit provided upstream of the print head is known as a configuration of a printing apparatus in which a print medium is transported to the print head (e.g., refer to JP-A-2013-230657).

(4) In order to improve a printing speed of the printing apparatus, for example, it is conceivable for the configuration disclosed in JP-A-2013-230657 to increase the size of the platen in the transport direction. With this configuration, the length of transport by which the print medium is intermittently transported toward the platen is increased. For this reason, the transport path from the steering unit to the platen needs to be lengthened, which leads to the problem that the print medium easily meanders. In addition, in order to suppress meandering of the print medium, the number of transport rollers needs to be increased to make the intervals between the transport rollers narrower, which increases the size and cost of the apparatus.

SUMMARY

(5) An aspect of the present disclosure is a printing apparatus including: a printing unit configured to perform printing on a print medium, a support unit disposed facing the printing unit, the support

unit supporting the print medium, a plurality of first transport rollers configured to transport the print medium to the support unit, a medium collection unit configured to wind in a roll shape the print medium on which printing was performed by the printing unit, and a plurality of second transport rollers configured to transport the print medium from the support unit to the medium collection unit, in which each interval between the plurality of first transport rollers is narrower than an average value of each interval between the plurality of second transport rollers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an overview of a printing apparatus according to the present embodiment.

(2) FIG. 2 is a side view illustrating a configuration of a first transport roller unit.

(3) FIG. 3 is a plan view of the main part of the first transport roller unit.

(4) FIG. 4 is an explanatory diagram of parallelism between first transport rollers.

DESCRIPTION OF EMBODIMENTS

1. Overall Configuration of Printing Apparatus

(5) Embodiments of the disclosure will be described below with reference to the accompanying drawings.

(6) FIG. 1 is a schematic diagram illustrating an overall configuration of a printing apparatus 1. In FIG. 1 and the drawings described below, XYZ orthogonal coordinates are illustrated in order to describe directions in a state in which the printing apparatus 1 is installed. The Z axis indicates a vertical direction in the state in which the printing apparatus 1 is installed, and can be referred to as a height direction. The X axis indicates a left-right direction of the printing apparatus 1. The Y axis is a direction orthogonal to the X axis, and can be referred to as a front-rear direction.

(7) The printing apparatus 1 performs printing on a print medium S with a printing unit 5. Various types of sheets can be used as the print medium S used in the printing apparatus 1. The following description introduces a configuration in which label paper formed by placing labels with an adhesive backing on release paper and winding the paper in a roll shape is used as the print medium S. Although a printing method of the printing apparatus 1 is not limited, in the embodiment, an inkjet printer that ejects ink to the print medium S with the printing unit 5 is described as an example of the printing apparatus 1.

(8) The printing apparatus 1 has a configuration in which a medium supply unit 3, the printing unit 5, a drying unit 7, a medium collection unit 8, and a control device 100 are disposed in a housing 2. The housing 2 is attached to a frame of the printing apparatus 1, which is not illustrated, and accommodates the above-described units.

(9) The print medium S is transported between the medium supply unit 3 and the medium collection unit 8, and a transport path of the print medium S is indicated by reference sign 10. A plurality of transport rollers 301 to 343 that come into contact with the print medium S are disposed in the transport path 10. The print medium S supplied from the medium supply unit 3 is transported in the transport direction indicated by the arrow F in the drawing in the transport path 10. The transport rollers 301 to 321 are positioned upstream of the printing unit 5 in the transport path 10.

On the other hand, the transport rollers 322 to 343 are positioned downstream of the printing unit 5.

(10) The printing apparatus 1 includes motors M1 to M6, and the motors M1 to M6 generate a driving force for transporting the print medium S. The feeding motor M1 drives the supply shaft 31 of the medium supply unit 3, the feeding nip motor M2 drives the transport roller 305, and the supply nip motor M3 drives the transport roller 318. These motors M1, M2, and M3 are for supply-side power for supplying the print medium S from the medium supply unit 3 to the printing unit 5. On the other hand, a discharge nip motor M4 drives the transport roller 334, the winding nip motor

M5 drives the winding transport roller **342**, and the winding motor M6 drives a winding shaft **81** of the medium collection unit **8**. These motors M4, M5, and M6 are for collection-side power for collecting the print medium S on which printing was performed by the printing unit **5** to the medium collection unit **8**.

(11) The medium supply unit **3** includes the cylindrical or columnar supply shaft **31**. A roll body **32** on which the print medium S is wound in a roll shape can be mounted on the supply shaft **31**, and the medium supply unit **3** supports the roll body **32** with the supply shaft **31**. The supply shaft **31** is rotated by the power of the motor M1 to feed the print medium S from the roll body **32** toward the printing unit **5**.

(12) The printing unit **5** prints characters and images on the label of the print medium S by ejecting ink toward the print medium S. The printing unit **5** includes a platen **54**, a plurality of ejection heads **51** that eject ink, and a carriage **53** that holds the ejection heads **51**. The platen **54** includes, for example, a rectangular flat surface and is parallel to the XY plane in the installation state of the printing apparatus **1**. The platen **54** supports the print medium S from below. A suction hole or the like for holding the print medium S on the platen **54** by applying suction force to the print medium S may be formed in the platen **54**. The platen **54** corresponds to an example of a support part.

(13) The carriage **53** is disposed above the platen **54** to face the platen **54**. The carriage **53** is capable of reciprocating in the X-axis direction along a first guide rail **25** installed in the printing apparatus **1** in the X-axis direction. The carriage **53** is capable of reciprocating in the Y-axis direction along a second guide rail, which is not illustrated, installed in the Y-axis direction. The printing apparatus **1** includes a first carriage motor M7 that moves the carriage **53** along the first guide rail **25** and a second carriage motor M8 that moves the carriage **53** along the second guide rail. The printing apparatus **1** moves the ejection head **51** in the X-axis direction and the Y-axis direction with power of the motors M7 and M8. With this configuration, the ejection head **51** can move over the print medium S supported by the platen **54** in the X-axis direction and the Y-axis direction to eject ink to the entire print medium S.

(14) When printing is performed on the print medium S, the printing apparatus **1** intermittently transports the print medium S. That is, the printing apparatus **1** transports the print medium S to the platen **54** and stops the transport of the print medium S in a state in which the print medium S is disposed in a predetermined region of the platen **54**. Here, the printing unit **5** causes the ejection head **51** to eject ink while moving the ejection head **51** in the X-axis direction and the Y-axis direction to perform printing on the print medium S. After printing, the printing apparatus **1** transports the print medium S until the printed portion of the print medium S moves downstream of the platen **54**. Then, the printing apparatus **1** stops the transport of the print medium S and performs printing with the printing unit **5**.

(15) A feeding-side buffer unit **60** and a meandering correction unit **70** that corrects meandering of the print medium S are disposed between the medium supply unit **3** and the printing unit **5** in the transport path **10**.

(16) The feeding-side buffer unit **60** includes a feeding-side fixed plate **65** that is directly or indirectly fixed to the frame of the printing apparatus **1** and a feeding-side movable plate **67** that can move up and down. The seven transport rollers **303**, **304**, **305**, **306**, **307**, **309**, and **311** are rotatably supported by the feeding-side fixed plate **65**. The two transport rollers **308** and **310** are rotatably supported by the feeding-side movable plate **67**.

(17) In the feeding-side buffer unit **60**, the print medium S is wound around each of the transport rollers **303** to **311**. The feeding-side buffer unit **60** applies tension to the print medium S in the direction of gravity due to the weights of the feeding-side movable plate **67** and the transport rollers **308** and **310** attached to the feeding-side movable plate **67**.

(18) When slack occurs in the print medium S between the medium supply unit **3** and the platen **54**, the feeding-side movable plate **67** is lowered by an amount corresponding to the slack of the print medium S, thereby absorbing the slack of the print medium S. In addition, when strong tension is

applied to the print medium S in the transport direction between the medium supply unit **3** and the platen **54**, the print medium S is fed from the feeding-side buffer unit **60** by the feeding-side movable plate **67** being raised by the tension. In this way, the feeding-side buffer unit **60** absorbs or alleviates the extra portion of the print medium S and excessive tension attributable to a shortage of the print medium S between the medium supply unit **3** and the platen **54**.

(19) The meandering correction unit **70** includes transport rollers **312** and **313**, and corrects meandering of the print medium S between the transport rollers **312** and **313**. Meandering of the print medium S is a phenomenon in which the transport direction of the print medium S is inclined from the transport path **10** to a direction along the Y axis. An ideal transport state is, for example, a state in which the print medium S is transported from the medium supply unit **3** to the platen **54** without moving in a direction along the Y axis. In this state, the print medium S fed out from the medium supply unit **3** moves along the X axis and reaches the platen **54**, without moving in a direction along the Y axis. However, the print medium S may be inclined with respect to the X axis due to the influence of an inclination of the transport rollers **303** to **311**, or the like. The state of such an inclination is called meandering. When the movement direction of the print medium S is a direction in which the print medium is inclined with respect to the X axis, the meandering correction unit **70** corrects the movement direction. The print medium S transported from the meandering correction unit **70** toward the platen **54** is in a state in which meandering is substantially eliminated.

(20) A meandering accuracy maintenance region **75** for maintaining the meandering accuracy of the print medium S is provided between the meandering correction unit **70** and the platen **54**. The meandering accuracy indicates a degree of meandering of the print medium S, that is, a degree of movement thereof in a direction intersecting the X axis, and may be referred to as transport accuracy. The transport rollers **313**, **314**, **315**, **316**, **317**, **318**, **319**, **320**, and **321** are disposed in the meandering accuracy maintenance region **75**.

(21) The transport rollers **313** and **314** are directly or indirectly attached to the frame of the printing apparatus **1**. The transport rollers **313** and **314** are rotatably supported.

(22) A vertical plate **71** fixed directly or indirectly to the frame of the printing apparatus **1** is disposed in the meandering accuracy maintenance region **75**. The transport roller **315** is rotatably supported at a lower end part of the vertical plate **71**. In addition, the six transport rollers **316**, **317**, **318**, **319**, **320**, and **321** are disposed at the central and upper parts of the vertical plate **71**. Each of the transport rollers **316** to **321** is rotatably supported by the vertical plate **71**. In the meandering accuracy maintenance region **75**, tension is applied to the print medium S being transported. This tension prevents the print medium S from wrinkling or sagging. The print medium S is transported to the platen **54** through the meandering accuracy maintenance region **75**. A configuration including the first transport rollers **313** to **321** and the vertical plate **71** disposed in the meandering accuracy maintenance region **75** is referred to as a first transport roller unit **4**. A configuration of the first transport roller unit **4** will be described below.

(23) The transport rollers **313** to **321** disposed in the meandering accuracy maintenance region **75** transport the print medium S while maintaining the meandering accuracy corrected by the meandering correction unit **70**.

(24) The drying unit **7** is disposed downstream of the printing unit **5** in the transport path **10**. Transport rollers **322**, **323**, **324**, **325**, and **326** are disposed in the drying unit **7**. The drying unit **7** includes a heater that heats the printing surface and/or the back of the printing surface of the print medium S, and dries the ink ejected by the printing unit **5**.

(25) Transport rollers **327**, **328**, **329**, and **330** are disposed downstream of the drying unit **7** in the transport path **10**. With these transport rollers **327** to **330**, the print medium S is disposed in a winding-side buffer unit **61**. The winding-side buffer unit **61** includes transport rollers **331** to **339**. The winding-side buffer unit **61** includes a winding-side fixed plate **68** fixed to a fixing unit, and a winding-side movable plate **69** that can move up and down. The six transport rollers **332** to **336** and

338 are rotatably supported on the winding-side fixed plate **68**, and one transport roller (dancer roller or winding-side movable roller) **337** is rotatably supported on the winding-side movable plate **69**. The transport rollers **339** to **343** are rotatably supported by a fixing unit.

(26) The print medium **S** is wound around each of the transport rollers **331** to **339** in the winding-side buffer unit **61**. The winding-side buffer unit **61** applies tension to the print medium **S** in the direction of gravity by using the weights of the winding-side movable plate **69** and the winding-side movable roller **337** attached to the winding-side movable plate **69**.

(27) When slack occurs in the print medium **S** between the drying unit **7** and the medium collection unit **8**, the winding-side movable plate **69** is lowered by the amount of the slack of the print medium **S** to absorb the slack of the print medium **S**. In addition, when strong tension is applied to the print medium **S** in the transport direction between the drying unit **7** and the medium collection unit **8**, the winding-side movable plate **69** is raised due to the tension, and thus the print medium **S** is fed out from the winding-side buffer unit **61**. In this way, the winding-side buffer unit **61** absorbs or alleviates the extra portion of the print medium **S** and excessive tension attributable to a shortage of the print medium **S** between the drying unit **7** and the medium collection unit **8**.

2. Configuration of First Transport Roller Unit

(28) FIG. **2** is a side view illustrating a configuration of the first transport roller unit **4**.

(29) Hereinafter, the plurality of transport rollers **313** to **321** that transport the print medium **S** from the meandering correction unit **70** to the printing unit **5** are referred to as first transport rollers **313** to **321**. In addition, the plurality of transport rollers **322** to **332** that transport the print medium **S** from the printing unit **5** to the winding-side buffer unit **61** are defined as second transport rollers **322** to **332**.

(30) A fixing unit **313A** that rotatably supports the transport roller **313** and a fixing unit **314A** that rotatably supports the transport roller **314** are disposed in the first transport roller unit **4**. The fixing units **313A** and **314A** and the vertical plate **71** are directly or indirectly fixed to the frame of the printing apparatus **1**.

(31) The vertical plate **71** includes a lower plate **71A** and an upper plate **71B**. The lower plate **71A** rotatably supports the transport roller **315**. The upper plate **71B** rotatably supports the first transport rollers **316**, **317**, **319**, **320**, and **321**. The first transport roller **318** is a driving roller driven by the supply nip motor **M3** as described above. The first transport roller **318** is rotatably supported by the upper plate **71B** and is coupled to the supply nip motor **M3**.

(32) The print medium **S** meanders due to factors such as an inclination of the rollers that transport the print medium **S** as described above. Specifically, factors that cause a deviation in transport and reduce the meandering accuracy are greatly affected by parallelism or alignment between the rollers, and a distance between the transport rollers. In other words, if the first transport rollers **313** to **321** are attached in parallel at high accuracy and the intervals between the first transport rollers **313** to **321** are short, the meandering accuracy of the print medium in the first transport roller unit **4** does not decrease. That is, the meandering accuracy corrected by the meandering correction unit **70** is maintained even in the first transport roller unit **4**.

(33) FIG. **3** is a plan view of the main part of the first transport roller unit **4** viewed from the direction **III** of FIG. **2**.

(34) The first transport roller **320** and the first transport roller **321** are disposed parallel to each other at a predetermined interval as illustrated in FIG. **3**. That is, the first transport rollers **320** and **321** are installed at the upper plate **71B** such that a rotation axis **320A** of the first transport roller **320** and a rotation axis **321A** of the first transport roller **321** have predetermined parallelism to be described later. In addition, an interval between the rotation axis **320A** and the rotation axis **321A** is referred to as an inter-axis distance. The inter-axis distance between the rotation axis **320A** and the rotation axis **321A** is equal to or less than a predetermined interval described later.

(35) FIG. **4** is an explanatory diagram of parallelism between the first transport rollers.

(36) With respect to the rotation axes of transport rollers, reference sign **A** indicates the rotation

axis of any one of the first transport rollers **313**, **314**, **315**, **316**, **317**, **318**, **319**, **320**, and **321**.

Reference sign B, C, D, and E are rotation axes of the first transport rollers adjacent to the first transport roller indicated by reference sign A. For example, reference sign A indicates the rotation axis **320A**, and reference signs B to E each indicate the rotation axis **321A**.

(37) Reference sign B indicates a state in which the parallelism between two first transport rollers is 0 mm. Reference sign C indicates a state in which the parallelism between two first transport rollers is within 0.15 mm. Reference sign D indicates a state in which the parallelism between two first transport rollers is within 0.15 mm. Reference sign E indicates an example in which the parallelism between two first transport rollers exceeds 0.15 mm in an inclined manner.

(38) Through a demonstration experiment, the present inventors have found that meandering of the print medium S can be alleviated to a preferable range by keeping the parallelism between adjacent first transport rollers of the first transport rollers **313** to **321** of the first transport roller unit **4** within 0.15 mm.

(39) In addition, the present inventors have found that, for the first transport rollers **313** to **321** of the first transport roller unit **4**, meandering of the print medium S can be alleviated to a preferable range when the inter-axis distance between adjacent first transport rollers is equal to or less than 400 mm.

(40) In these cases, the first transport rollers **313** to **321** have a diameter of 80 mm, and desirably a diameter of 10 mm to 100 mm. Note that the present disclosure is not applied to thin transport rollers such as those having a diameter less than 10 mm.

(41) Therefore, the present inventors have found the following about at least one first transport roller and a first transport roller adjacent thereto of the first transport rollers **313** to **321** of the first transport roller unit **4**. That is, at least one of the following two conditions that the parallelism between these two first transport rollers is within 0.15 mm or that the inter-axis distance between these two first transport rollers is equal to or less than 400 mm may be satisfied. In addition, both of these two conditions may be satisfied.

(42) Furthermore, for all of the first transport rollers **313** to **321**, at least one of the conditions that the parallelism with respect to any adjacent first transport rollers satisfies the above condition or that the inter-axis distance with respect to any adjacent first transport rollers satisfies the above condition may be satisfied. In addition, for all of the first transport rollers **313** to **321**, the parallelism and the inter-axis distance with respect to any adjacent first transport rollers may satisfy the above-described conditions.

(43) Note that the parallelism may not be parallelism between adjacent first transport rollers, but may be parallelism with respect to a reference transport roller with any of the first transport rollers **313** to **321** serving as the reference transport roller. In this case, deviation in parallelism gradually increases as the number of transport rollers increases. For this reason, it is desirable that the parallelism between adjacent transport rollers be set to be in a range narrower than 0.15 mm.

(44) For example, with one first transport roller serving as a reference, if the rotation axis B of another transport roller is parallel to the reference rotation axis A, the parallelism is 0 mm. The parallelism of the rotation axes C and D of other transport rollers is within α mm. The parallelism of the rotation axis E is outside the range of α mm. In this embodiment, with the rotation axis of the first transport roller **321** positioned immediately before the platen **54** serving as a reference rotation axis A, the parallelism of the first transport rollers **313** to **321** is set to be within α mm ($\alpha=0.15$). It has been verified that if the parallelism of the first transport rollers **313** to **321** is set to be within 0.15 mm, the meandering accuracy of the print media S from the meandering correction unit **70** to the printing unit **5** is favorably maintained.

(45) For example, it is assumed that with the rotation axis **321A** of the first transport roller **321** serving as a reference indicated by reference sign A, the parallelism of the first transport rollers **313** to **321** is set to be within 0.15 mm. In this case, for example, a state may occur in which, of two adjacent first transport rollers, the parallelism of one first transport roller with respect to the

reference is 0.15 mm, the parallelism of the other first transport roller with respect to the reference is -0.15 mm, and thus the one first transport roller and the other first transport roller are inclined in opposite directions. In this state, the parallelism between the adjacent first transport rollers is not within 0.15 mm.

(46) In another embodiment, with the rotation axis of the first transport roller **321** serving as a reference rotation axis A, the parallelism of the first transport rollers **313** to **321** with respect to the reference is set to be within 0.15 mm, and furthermore the parallelism between the adjacent first transport rollers is set to be within 0.15 mm. According to this configuration, the parallelism of the first transport rollers **313** to **321** can be maintained at good accuracy, and the meandering accuracy of the print medium S from the meandering correction unit **70** to the printing unit **5** is more favorably maintained.

(47) In addition, for example, of the first transport rollers **313** to **321**, the first transport roller **321** positioned immediately before the platen **54** is set as the reference rotation axis A. In this case, the parallelism of the rotation axis of each of the first transport rollers **313** to **321** is set to be within 0.15 mm, and the inter-axis distance between the first transport rollers **313** to **321** is set to be equal to or less than 400 mm. In this example, even if the inter-axis distance between the first transport rollers **313** to **321** is set to be relatively long in a range equal to or less than 400 mm, the amount of change in deviation of the print media S in the transport direction is small, meandering is alleviated due to the influence of the parallelism of the next transport roller, and thus the meandering accuracy can be improved.

(48) Although the printing apparatus **1** includes the meandering correction unit **70** in the embodiment described above, the configuration described above can also be applied to a printing apparatus **1** not including the meandering correction unit **70**. In this case, a plurality of transport rollers that transport the print media S from the medium supply unit **3** to the printing unit **5** may be set as the first transport rollers, and the parallelism of the rotation axis of each of the first transport rollers may be set to be within 0.15 mm. In addition, the inter-axis distance between the plurality of first transport rollers may be set to be equal to or less than 400 mm.

(49) In addition, in this embodiment, the interval between two adjacent first transport rollers of the first transport rollers **313** to **321** is set to be narrower than the intervals between the second transport rollers **322** to **332**. The average value of the intervals between the first transport rollers **313** to **321** may be set to be narrower than the average value of the intervals between the second transport rollers **322** to **332**.

(50) Since the second transport rollers **322** to **332** transport the print medium S on which printing was performed by the printing unit **5**, the print quality is little affected by a decrease in the meandering accuracy in the second transport rollers **322** to **332**. In addition, by increasing the inter-axis distance between the second transport rollers **322** to **332**, the number of rollers downstream of the printing unit **5** can be reduced, as well as reduction in size and weight of the printing apparatus **1**, reduction in manufacturing costs of the printing apparatus **1**, and reduction in the number of manufacturing steps of the printing apparatus **1** can be achieved.

(51) In addition, when the platen **54** of the printing apparatus **1** is increased in size in order to improve the printing speed, an increase in the number of transport rollers in the entire printing apparatus **1** can be avoided even if the transport path **10** of the print medium S from the medium supply unit **3** to the printing unit **5** is lengthened.

3. Other Embodiments

(52) The above embodiment is merely a specific example to which the present disclosure is applied. The present disclosure is not limited to the configuration of the above embodiment, and can be implemented in various aspects without departing from the gist of the disclosure.

(53) For example, although the printing apparatus **1** has been described in the above-described embodiment as having a configuration in which the feeding-side buffer unit **60**, the winding-side buffer unit **61**, and the drying unit **7** are included, the disclosure is also applicable to a printing

apparatus **1** having no such configuration. In addition, the printing apparatus **1** is not limited to an apparatus in which a continuous sheet wound around the roll body **32** is used as the print medium **S**, and an aspect of the continuous sheet can be changed as desired.

(54) In addition, although the ejection head **51** in the above embodiment is configured to move in the X-axis direction and the Y-axis direction on the print medium **S** supported by the platen **54** and eject ink to the entire print medium **S**, a so-called single-pass printing method may be adopted in which the ejection head **51** is longer than the print medium **S** in the width direction, i.e., in the Y-axis direction, and in which an image is formed by moving the ejection head **51** only once in the X-axis direction.

(55) In addition, although the inkjet printing apparatus **1** has been described in the above-described embodiment, this is an example, and the disclosure can be applied to printing apparatuses adopting other printing methods.

(56) In addition, the numbers of transport rollers, first transport rollers, and second transport rollers, and the number of motors that drive the transport rollers described in the above embodiment are merely examples, and can be arbitrarily changed.

4. Configurations Described Based on Embodiments

(57) The following configurations are described based on the above embodiment.

(58) Configuration 1

(59) A printing apparatus including: a printing unit that performs printing on a print medium; a support unit disposed facing the printing unit, the support unit supporting the print medium; a plurality of first transport rollers that transport the print medium to the support unit; a medium collection unit that winds in a roll shape the print medium on which printing was performed by the printing unit; and a plurality of second transport rollers that transport the print medium from the support unit to the medium collection unit, in which each interval between the plurality of first transport rollers is narrower than an average value of each interval between the plurality of second transport rollers.

(60) According to the printing apparatus described in Configuration 1, meandering accuracy of the print medium in the first transport rollers that transport the print medium to the printing unit can be made favorable. In addition, since each interval between the second transport rollers can be made larger than each interval between the first transport rollers, increase in the number of transport rollers can be curbed. With this configuration, reduction in size and weight of the printing apparatus, reduction in manufacturing costs of the printing apparatus, and reduction in the number of manufacturing steps of the printing apparatus can be achieved.

(61) Configuration 2

(62) The printing apparatus according to Configuration 1, in which an interval between adjacent first transport rollers of the plurality of first transport rollers is equal to or less than 400 mm.

(63) According to the printing apparatus of Configuration 2, by setting the interval between the first transport rollers to be equal to or less than 400 mm, the meandering accuracy of the print medium upstream of the printing unit can be made favorable and the print quality can be improved.

(64) Configuration 3

(65) The printing apparatus according to Configuration 1 or 2, in which parallelism of the first transport rollers included in the plurality of first transport rollers with respect to a reference transport roller is equal to or less than 0.15 mm.

(66) According to the printing apparatus of Configuration 3, by setting the parallelism of the first transport rollers with respect to a reference transport roller to be equal to or less than 0.15 mm, the meandering accuracy of the print medium upstream of the printing unit can be made favorable and the print quality can be improved.

(67) Configuration 4

(68) The printing apparatus according to Configuration 3, in which the reference transport roller is the first transport roller closest to the support unit of the plurality of first transport rollers.

(69) According to the printing apparatus described in Configuration 4, with the first transport roller positioned closest to the printing unit serving as a reference, the parallelism of the first transport rollers is set to be equal to or less than 0.15 mm, and thus the meandering accuracy of the print medium at a position at which the print quality is greatly affected can be made favorable.

(70) Configuration 5

(71) The printing apparatus according to any one of Configurations 1 to 4, in which the parallelism between adjacent first transport rollers of the plurality of first transport rollers is equal to or less than 0.15 mm.

(72) According to the printing apparatus of Configuration 5, by setting the parallelism between adjacent first transport rollers to be equal to or less than 0.15 mm, the meandering accuracy of the print medium upstream of the printing unit can be made favorable and the print quality can be improved.

(73) Configuration 6

(74) The printing apparatus according to any one of Configurations 1 to 5, further including a meandering correction unit upstream of the support unit, in which the plurality of first transport rollers are disposed between the meandering correction unit and the support unit in a transport path of the print medium.

(75) According to the printing apparatus described in Configuration 6, when the print medium whose meandering accuracy has been corrected by the meandering correction unit is transported, the meandering accuracy can be made favorable.

Claims

1. A printing apparatus comprising: a printing unit configured to perform printing on a print medium; a support unit disposed facing the printing unit, the support unit supporting the print medium; a plurality of first transport rollers configured to transport the print medium to the support unit; a medium collection unit configured to wind in a roll shape the print medium on which printing was performed by the printing unit; and a plurality of second transport rollers configured to transport the print medium from the support unit to the medium collection unit, wherein each interval between adjacent rollers of the plurality of first transport rollers is narrower than an average value of each interval between adjacent rollers of the plurality of second transport rollers.
 2. The printing apparatus according to claim 1, wherein all of an interval between adjacent first transport rollers of the plurality of first transport rollers is equal to or less than 400 mm.
 3. The printing apparatus according to claim 1, wherein parallelism of the first transport rollers included in the plurality of first transport rollers with respect to a reference transport roller is equal to or less than 0.15 mm.
 4. The printing apparatus according to claim 3, wherein the reference transport roller is the first transport roller closest to the support unit of the plurality of first transport rollers.
 5. The printing apparatus according to claim 3, wherein all of a parallelism between adjacent first transport rollers of the plurality of first transport rollers is equal to or less than 0.15 mm.
 6. The printing apparatus according to claim 1, further comprising a meandering correction unit upstream of the support unit, wherein the plurality of first transport rollers are disposed between the meandering correction unit and the support unit in a transport path of the print medium.
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